



MINING DEALER BEST PRACTICE

Blade Edge Wear Levels Monitoring Using a Color Gauge on 16H/16M/16 Motor Grader

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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1.0 Introduction

Currently, the monitoring of the levels of wear on the blade edges is carried out by the maintenance team, in which it is necessary to travel to the place of operation and request the equipment to stop to perform the measurement.

These downtimes impact equipment productivity, use of resources and labor.

The idea is that the operator uses the Color Gauge before the start of his work shift to check the wear range of the blade edge and, thus, communicate the maintenance, providing a follow-up of the edge wear.

The objective of this Best Practice is to assist the operators in monitoring the blade wear edges, creating, through a Color Gauge, bands with colors, which will guide the operator to inform maintenance team about the current level of edge wear. This tool will also allow the operator to know if the edges of the blade are in proper condition for operation.

2.0 Best Practice Description

This Best Practice proposes the elaboration of a Color Gauge with three colors that will classify the wear levels into three bands. These tracks have the following categories:

Green Band: it shows that the edge measurement is within the work specifications.

Yellow Band: it indicates that the edge is close to the minimum range limit specified for work and, in this case, maintenance technicians must schedule its replacement as soon as possible.

Red Band: it guides the operator and maintenance team that the edge has already exceeded the specified work wear limit and it must be replaced immediately.

In this way, before starting the service, using this Color Gauge, the operator can measure the edge wear level and inform the maintenance team.

Maintenance, in turn, through the information shared by the operator, can monitor the edge wear before the start of each work shift.

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Picture 1 - Operator using gauge color to assess blade edge wear level



Picture 2 - Gauge color for 16M motor grader

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3.0 Implementation Steps

- 1) Color Gauge fabrication with a 1.5mm thick sheet metal and specific dimensions for the edge as shown below:



Picture 3 - Color Gauge dimensions for manufacturing

16' Blades

Machine Models: 160H, 160M, 14H, 14M, 16H, 16M, 160, 16

Profile	Type	Width Tooth & Gap (mm)	Length	Qty	Hardware	Moldboard Cutting Edges			
						19mm	25mm	35mm	45mm
Flat	Standard	8"	8'	2	17	4T-3037	7T-3495	4T-8317	104-3628
		10"				4T-8511	132-1086		
		11"				140-6832	4Z-9188		
		12"							
		13"							
	Std. A.R.M.	10"				195-7177		128-2577	
	13"	489-6706		554-0845					
							554-0846		

4T-8317

Height 10"

Color Gauge information

Green = 110 mm (=43,31%)

Yellow = 20 mm (7,87%)

Picture 4 - Standard measurement references for 16M, 16H and 16

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- 2) Guide operators how to use the Color Gauge.
- 3) Guide maintenance team on how to use the Color Gauge.
- 4) Establish the measure collection frequency and an information flow and between operator and maintenance departments.

4.0 Benefits

- 1) Time Saving
The use of the Color Gauge will reduce machine downtime for edge measurement, as the operator itself evaluates the edges before starting the operation.
- 2) Efficiency Improvement
Color Gauge will allow a better control over edge wear monitoring and changing edges improved predictability.
- 3) Resource savings
The use of Color Gauge to measure the edges will prevent the maintenance team from moving towards the equipment on a daily basis and will also prevent damage to the edge fixing base due to lack of wear monitoring.

5.0 Resources Required

- 1) Color Gauge - Supply the color gauge as per the specified colors and measurements.
- 2) Operators – train operators how to use the Color Gauge efficiently.
- 3) Maintenance team - Train the maintenance team in managing the flow of information about the measures passed on by the operator.

The 4T-8317 edge is the recommended option for MG's 16H, 16M, 16 and, for this reason, the gauge measurements are based on the specified wear parameters for it.

6.0 Supporting Attachments / References

Motor Grader Cutting Edge Options – www.Dealer.cat.com .
CMIC - PECJ0025-03 GET PSK Section 10 - Motor Graders - www.Dealer.cat.com .

7.0 Related Best Practices

N/A

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